

## KILLEEN SKYLARK, TX, USA

WMO: 722575

Lat: 31.083N

Lon: 97.683W

Elev: 841

StdP: 14.25

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03972

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 25.6       | 28.7 | 8.9                           | 8.9 | 33.1 | 13.8 | 11.4 | 37.2 | 25.3                  | 56.5 | 23.0 | 56.4 | 8.5                   | 0    | 0.472 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 21.3                   | 100.1           | 74.2 | 99.0 | 74.3 | 96.9 | 74.5 | 78.0                | 91.9 | 77.3 | 91.1 | 76.6 | 90.0 | 9.7                  | 160  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 74.4                            | 132.7 | 82.5 | 73.4 | 128.1 | 82.0 | 72.9 | 125.8 | 81.9 | 41.9          | 92.2 | 41.1 | 91.1 | 40.5 | 89.3 | 83.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |       |                    | n-Year Return Period Values of Extreme Temperature |           |       |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|------|-------|--------------------|----------------------------------------------------|-----------|-------|------------|-------|------------|-------|------------|-------|
| 1%                |      | 2.5% | 5%                         | Mean |       | Standard Deviation |                                                    | n=5 years |       | n=10 years |       | n=20 years |       | n=50 years |       |
| Min               | Max  | Min  | Max                        | Min  | Max   | Min                | Max                                                | Min       | Max   | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)   | (h)                | (i)                                                | (j)       | (k)   | (l)        | (m)   | (n)        | (o)   | (p)        |       |
| 21.9              | 19.7 | 18.0 | DB                         | 20.7 | 103.2 | 3.2                | 2.3                                                | 18.4      | 104.8 | 16.6       | 106.1 | 14.8       | 107.4 | 12.4       | 109.1 |
|                   |      |      | WB                         | 17.9 | 79.6  | 3.3                | 1.2                                                | 15.6      | 80.5  | 13.6       | 81.2  | 11.8       | 81.8  | 9.4        | 82.7  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 67.9   | 49.6  | 53.3  | 59.9  | 67.5  | 75.4  | 81.7  | 84.7  | 85.0  | 78.8  | 69.2  | 58.6  | 50.4 |
|                                                                                  | DBStd       | 14.85  | 9.79  | 10.56 | 9.33  | 7.32  | 5.95  | 3.85  | 3.39  | 3.67  | 5.96  | 7.80  | 9.31  | 9.64 |
|                                                                                  | HDD50       | 376    | 128   | 78    | 26    | 1     | 0     | 0     | 0     | 0     | 0     | 1     | 29    | 113  |
|                                                                                  | HDD65       | 1847   | 485   | 346   | 209   | 57    | 6     | 0     | 0     | 0     | 3     | 49    | 230   | 462  |
|                                                                                  | CDD50       | 6913   | 115   | 169   | 333   | 527   | 787   | 951   | 1075  | 1084  | 863   | 596   | 288   | 125  |
|                                                                                  | CDD65       | 2909   | 6     | 17    | 51    | 134   | 327   | 501   | 610   | 619   | 416   | 179   | 39    | 10   |
|                                                                                  | CDH74       | 32451  | 51    | 128   | 368   | 1096  | 3039  | 5733  | 7836  | 7999  | 4416  | 1485  | 248   | 52   |
|                                                                                  | CDH80       | 16100  | 6     | 29    | 69    | 321   | 1235  | 2869  | 4391  | 4505  | 2132  | 497   | 42    | 4    |
| Wind                                                                             | WSAvg       | 8.7    | 8.6   | 9.2   | 9.9   | 10.5  | 10.0  | 9.2   | 8.3   | 7.6   | 6.9   | 7.9   | 8.2   | 8.1  |
| Precipitation                                                                    | PrecAvg     | 35.1   | 2.4   | 2.5   | 3.1   | 2.8   | 4.4   | 3.6   | 2.1   | 2.3   | 3.2   | 3.7   | 2.9   | 2.6  |
|                                                                                  | PrecMax     | 47.7   | 5.6   | 7.4   | 9.1   | 5.9   | 10.7  | 9.7   | 6.1   | 8.0   | 8.4   | 11.0  | 10.1  | 8.9  |
|                                                                                  | PrecMin     | 21.7   | 0.3   | 0.1   | 0.0   | 0.5   | 1.1   | 0.5   | 0.0   | 0.1   | 0.4   | 0.4   | 0.3   | 0.2  |
|                                                                                  | PrecStd     | 7.9    | 1.5   | 1.8   | 1.9   | 1.6   | 2.3   | 2.2   | 1.6   | 2.3   | 2.1   | 2.7   | 2.5   | 2.2  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 80.5  | 83.9  | 85.7  | 91.5  | 96.7  | 100.1 | 102.4 | 102.5 | 101.6 | 91.4  | 84.3  | 80.2 |
|                                                                                  |             | MCWB   | 60.9  | 61.5  | 65.1  | 69.0  | 72.7  | 74.6  | 73.7  | 74.2  | 71.9  | 71.9  | 68.0  | 63.8 |
|                                                                                  | 2%          | DB     | 73.8  | 78.2  | 81.6  | 87.7  | 92.6  | 97.3  | 100.0 | 100.2 | 97.0  | 89.6  | 80.8  | 74.5 |
|                                                                                  |             | MCWB   | 59.7  | 61.6  | 64.7  | 68.6  | 73.4  | 75.0  | 74.1  | 74.4  | 73.4  | 71.8  | 66.2  | 62.0 |
|                                                                                  | 5%          | DB     | 70.2  | 73.2  | 78.8  | 83.6  | 90.1  | 94.6  | 98.7  | 99.1  | 93.2  | 85.6  | 76.7  | 70.5 |
|                                                                                  |             | MCWB   | 59.0  | 60.8  | 63.8  | 67.9  | 72.9  | 75.0  | 74.3  | 74.6  | 73.1  | 69.5  | 64.9  | 60.8 |
|                                                                                  | 10%         | DB     | 65.9  | 69.9  | 74.4  | 80.9  | 87.6  | 91.8  | 96.5  | 97.0  | 90.6  | 81.9  | 72.9  | 66.3 |
|                                                                                  |             | MCWB   | 57.6  | 60.6  | 62.4  | 66.5  | 72.6  | 74.7  | 74.7  | 74.8  | 72.8  | 68.0  | 63.7  | 58.6 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 66.7  | 68.6  | 70.4  | 74.4  | 77.8  | 79.1  | 78.8  | 78.5  | 78.1  | 76.2  | 72.5  | 68.9 |
|                                                                                  |             | MCDB   | 73.2  | 75.1  | 79.3  | 85.7  | 89.7  | 92.6  | 92.9  | 93.8  | 90.5  | 85.9  | 79.3  | 74.5 |
|                                                                                  | 2%          | WB     | 64.4  | 66.4  | 68.3  | 72.3  | 76.3  | 77.8  | 77.5  | 77.5  | 76.9  | 74.4  | 69.7  | 66.2 |
|                                                                                  |             | MCDB   | 69.7  | 72.0  | 76.3  | 82.7  | 87.2  | 91.1  | 92.0  | 92.6  | 89.0  | 83.3  | 76.4  | 71.5 |
|                                                                                  | 5%          | WB     | 61.9  | 64.5  | 66.9  | 70.7  | 75.0  | 76.9  | 76.9  | 76.9  | 75.7  | 73.0  | 67.6  | 63.4 |
|                                                                                  |             | MCDB   | 67.6  | 69.9  | 74.2  | 80.2  | 85.1  | 89.6  | 91.1  | 91.8  | 87.1  | 81.3  | 74.1  | 68.9 |
|                                                                                  | 10%         | WB     | 58.5  | 61.9  | 65.1  | 68.9  | 73.6  | 76.0  | 76.1  | 76.1  | 74.7  | 71.3  | 65.6  | 59.7 |
|                                                                                  |             | MCDB   | 65.1  | 68.8  | 71.6  | 77.0  | 83.5  | 87.9  | 89.9  | 90.6  | 85.6  | 78.9  | 71.8  | 65.2 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 21.2  | 20.5  | 21.2  | 21.4  | 19.4  | 19.6  | 21.3  | 21.3  | 20.7  | 20.7  | 20.3  | 20.5 |
|                                                                                  |             | MCDBR  | 27.3  | 25.0  | 24.3  | 24.8  | 22.4  | 22.8  | 24.5  | 24.5  | 23.7  | 23.0  | 23.2  | 24.7 |
|                                                                                  |             | MCWBR  | 16.8  | 15.2  | 13.0  | 12.5  | 8.8   | 6.9   | 6.5   | 6.8   | 7.7   | 10.5  | 13.2  | 16.2 |
|                                                                                  | 5% WB       | MCDBR  | 20.9  | 20.1  | 19.6  | 19.1  | 18.9  | 19.2  | 20.8  | 21.0  | 19.6  | 18.5  | 18.6  | 20.9 |
|                                                                                  |             | MCWBR  | 15.8  | 15.0  | 12.4  | 10.9  | 9.0   | 7.1   | 6.6   | 6.5   | 7.4   | 9.9   | 13.2  | 16.3 |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.314  | 0.329 | 0.355 | 0.395 | 0.430 | 0.424 | 0.437 | 0.428 | 0.398 | 0.369 | 0.344 | 0.318 |      |
|                                                                                  | taud        | 2.517  | 2.480 | 2.420 | 2.329 | 2.287 | 2.342 | 2.342 | 2.353 | 2.384 | 2.450 | 2.462 | 2.510 |      |
|                                                                                  | Ebn at Noon | 291    | 296   | 294   | 285   | 274   | 274   | 270   | 272   | 277   | 278   | 277   | 283   |      |
|                                                                                  | Edh at Noon | 28     | 31    | 36    | 41    | 43    | 40    | 40    | 39    | 36    | 32    | 29    | 26    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 955    | 1113  | 1443  | 1735  | 1841  | 2103  | 2135  | 1989  | 1655  | 1334  | 1006  | 843   |      |
|                                                                                  | RadStd      | 101    | 188   | 128   | 158   | 168   | 161   | 163   | 114   | 143   | 196   | 121   | 114   |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |     |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|-----|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |     |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   | (m) |
| Station Only            |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |     |
| Regional (45 neighbors) |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | +128  |     |

Nomenclature: See separate page